

Share Your Knowledge

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom

So if you have an interesting topic or original article to share, please let us know at efyeditorial@efy.in

We would love to hear your good ideas too!

electronics
FOR YOU

Leading manufacturers of industrial sensors

- Analog Devices
- Broadcom
- Digi International
- Honeywell International
- Infineon Technologies
- Libelium
- NXP Semiconductors
- Omron Corporation
- Robert Bosch GmbH
- ROHM Semiconductor
- Sensirion
- STMicroelectronics
- TE Connectivity
- Texas Instruments
- Vishay Intertechnology

satellite, IR, RF and Bluetooth. All industries are different and so are their exact requirements and applications. Hence various standards for sensor connectivity are defined by industrial alliances like HART and Zigbee to meet the requirements of various industrial applications. ZigBee operates at 2.4GHz with data rate of 250kbps. Depending on power output and environmental characteristics, outdoor line-of-sight range of these sensors may be up to 1500 metres. IoT application standards like IEEE 802.15.4 and Narrowband IoT (NB-IoT) enable a wide range of devices and applications.

Industrial applications of wireless sensors

Monitoring machinery and equipment. Wireless sensor networks are deployed to monitor machinery condition-based maintenance. Wireless sensors enable easy access to inaccessible locations, rotating machinery, hazardous or restricted areas, and mobile equipment, which otherwise are difficult to access using wired technology.

Data logging. Wireless sensors are also employed to collect data for environmental monitoring—for example, monitoring the temperature inside a fridge or monitoring the water level in the overflow tank of a nuclear power plant.

Water management. Wireless sensors are suitable for monitoring the quality of underground or surface water and preventing wastage

of water. Especially for remote areas having limited infrastructure, wireless sensor networks are ideal as these are fast to deploy and least intrusive for the local environment.

Chemical process monitoring. The concentration distribution of materials during chemical processing can be monitored and controlled using sensors networks. Individual sensor nodes can communicate with other nodes to provide an overall picture of the process. The information may include temperature, pressure, impedance, pH and dissolved gases.

Leading industrial sensor manufacturers

There are many manufacturers in the industrial sensors space, including a few from India. Some major semiconductor manufacturers like Texas Instruments, NXP Semiconductors, Analog Devices, STMicroelectronics and Vishay Intertechnology also manufacture sensors for various industrial applications. Companies like Honeywell International Inc. develop wireless monitoring and control products focused on industrial applications. ROHM Semiconductor offers a broad portfolio of industrial sensors including proximity sensors, light sensors, Hall sensors, capacitive and MEMS sensors. TE Connectivity offers different types of sensors for a wide range of industries including automotive, industrial, medical, appliance and defence.

Plug & Sense devices from Libelium are unique sensors that connect with the most common industrial communication protocols like RS-232, RS-485, Modbus, current loop and CAN bus. These are useful for a wide variety of applications including industrial sensing and automation, military, automobile and remote factory management.

To sum up

Different types of sensors are available for different industrial application requirements. Selecting the right sensor is important for various industrial process control applications. Smart wireless sensors are becoming popular to address various issues in modern industrial applications. **EFY**